

Advanced (Carolina Reaper)

Open Ended Questions (FRQ)

Q9. A coffee shop sells two types of coffee, Espresso and Cappuccino. The Espresso machine can produce at most 50 cups per hour, and the Cappuccino machine can produce at most 30 cups per hour. If the demand for Espresso is x cups per hour and the demand for Cappuccino is y cups per hour, and x and y must be integers, find the feasible region that represents the total production capacity of the coffee shop. Graph the region and provide the vertices.

Q10. Maya has 80 to spend on video games and snacks. She wants to buy x video games at 10 each and y snacks at 5 each. The store has a limit of 5 video games.

Chef's Tips

- Tip 1: Ensure students understand the context of each problem.
- Tip 2: Encourage students to graph feasible regions for systems of inequalities.
- Tip 3: Remind students to check for integer solutions when required.